

Operating Systems Questions 1–40 with Answers

Original question wording and answer choices kept as provided.

Question 1

Pages, sectors and page frames are of the same size to increase the _____ of the OS.

Select one:

- a. latency
- b. concurrency
- c. efficiency
- d. dependency

Answer: c. efficiency

Question 2

What are the 4 necessary conditions to be eliminated to avoid deadlock?

Select one:

- a. Race, Starvation, Circular wait and No preemption.
- b. Resource holding, No preemption, Livelock and Locking.
- c. Mutual exclusion, Resource holding, No preemption and Circular wait
- d. Livelock, Starvation, Mutual exclusion and Resource holding

Answer: c. Mutual exclusion, Resource holding, No preemption and Circular wait

Question 3

_____ is a situation where the resources needed by a job to finish execution are held by another job, which in turn, is waiting for other resources to become available

Select one:

- a. Aging
- b. Deadlock
- c. Starvation

Answer: b. Deadlock

Question 4

The memory allocation schemes eliminate the requirement that the entire program needs to reside in memory after execution.

Select one:

- True
- False

Answer: False

Question 5

Below are the example of possible cases of deadlock EXCEPT _____.

Select one:

- a. Deadlocks in spooling
- b. Deadlocks in dedicated device allocation
- c. Deadlocks on file requests
- d. Deadlocks in database allocation

Answer: d. Deadlocks in database allocation

Question 6

What are the types of Operating System?

Select one:

- a. Batch, Interactive, Real-time and Hybrid systems.
- b. Main memory, disk controller, serial and parallel interface.
- c. CPU, Main memory, I/O devices and kernel.
- d. ALU, registers, internal control and bus control.

Answer: a. Batch, Interactive, Real-time and Hybrid systems.

Question 7

One processor can be shared by several jobs/processes if and only if the OS has a _____ and _____ to determine when to stop working on one job and proceed to another.

Select one:

- a. scheduling policy, scheduling algorithm
- b. pages, algorithm
- c. frames and segments
- d. memory allocation scheme, algorithm

Answer: a. scheduling policy, scheduling algorithm

Question 8

Below are the responsibilities of a Process scheduler EXCEPT _____.

Select one:

- a. Decides when a job should be interrupted
- b. Determines a job has concluded and should be terminated
- c. Determines which queue a job should be moved to during its execution
- d. Decides when a job should be stopped.

Answer: d. Decides when a job should be stopped.

Question 9

Below are the differences between paging and segmentation EXCEPT _____.

Select one:

- a. Pages are physical units.
- b. Segments are invisible to users and of variable sizes.
- c. Segments are logical units.
- d. Pages are invisible to users and fixed sizes.

Answer: b. Segments are invisible to users and of variable sizes.

Question 10

Which of the following is NOT TRUE about recovery methods?

Select one:

- a. Stops all jobs from entering the system.
- b. Identify jobs involved and terminate one at a time.
- c. Terminate all active jobs and restart them.
- d. Terminate only jobs involved in a deadlock and resubmit.

Answer: a. Stops all jobs from entering the system.

Question 11

When the system is overloaded, the _____ removes active jobs from memory to reduce the degree of multi-programming and allows jobs to be completed faster.

Select one:

- a. middle-level scheduler
- b. low-level scheduler
- c. high-level scheduler

Answer: a. middle-level scheduler

Question 12

“The Dining Philosophers” problem introduced by Dijkstra is trying to solve the problem of ____.

Select one:

- a. Livelock
- b. Starvation
- c. Race
- d. Deadlock

Answer: b. Starvation

Question 13

Fixed partition scheme fully utilizes memory when the first jobs are loaded.

Select one:

- True
- False

Answer: False

Question 14

This memory allocation scheme overcomes the problems of segmentations such as compaction, external fragmentation and secondary storage handling.

Select one:

- a. Segmented/Demand Paged
- b. Paged allocation
- c. Demand Paging
- d. Segmented allocation

Answer: a. Segmented/Demand Paged

Question 15

The Banker's Algorithm proposed by Dijkstra regulates _____ to avoid deadlocks.

Select one:

- a. process
- b. synchronized process
- c. resource allocation
- d. file allocation

Answer: c. resource allocation

Question 16

Since segment sizes may vary in Segmented Memory Allocation Scheme, the displacement must be _____ to make sure it's not out of the segment's range.

Select one:

- a. verified
- b. rearrange
- c. calculated
- d. executed

Answer: a. verified

Question 17

What is the advantage of storing programs in non-contiguous locations?

Select one:

- a. Memory is used more efficiently because an empty page frame can be used by many jobs.
- b. Memory is used more efficiently because page frame can be used by more pages of any job.
- c. Memory is used more efficiently because an empty page frame can be used by any page of any job.
- d. Memory is used more efficiently because an empty sector can be used by any page of any job.

Answer: c. Memory is used more efficiently because an empty page frame can be used by any page of any job.

Question 18

The overhead cost of running the _____ algorithm can be quite high when there are many active jobs and many devices because it has to be invoked for each request.

Select one:

- a. deadlock
- b. starvation
- c. scheduling
- d. avoidance

Answer: d. avoidance

Question 19

Multi-programming requires that the _____ be allocated to each job/process for a period of time and _____ when the period expires.

Select one:

- a. resource, swap out
- b. processor, deallocated
- c. register, deallocated
- d. memory, deleted

Answer: b. processor, deallocated

Question 20

Once a job is selected for _____, the _____ determines when each step (or set of steps) is executed (also based on certain criteria).

Select one:

- a. processing, processor
- b. processing, Associative register
- c. execution, Job Scheduler
- d. execution, Process Scheduler

Answer: c. execution, Job Scheduler

Question 21

In Paged Memory Allocation Scheme, there is no need for compaction. Why?

Select one:

- a. Locates enough empty page frames in main memory.
- b. There is no internal fragmentation in pages.
- c. There is no external fragmentation between frame pages.
- d. Loads the pages (not necessarily contiguously).

Answer: c. There is no external fragmentation between frame pages.

Question 22

The shifting and setting the leftmost bit when a page is referenced in LRU gives a _____ of page usage.

Select one:

- a. feedback
- b. value
- c. history
- d. choice

Answer: c. history

Question 23

Absolute address calculated using segment number and displacement refers to _____.

Select one:

- a. Associative memory
- b. Virtual Memory with Segmentation

- c. Working Set
- d. Virtual Memory with Paging

Answer: b. Virtual Memory with Segmentation

Question 24

A thread is a unit smaller than a process.

Select one:

- True
- False

Answer: True

Question 25

In Demand Paging Allocation Scheme, Page Map Table contains info to determine if the page contents have been modified or not. Why?

Select one:

- a. To determine whether it need to be rewritten to secondary storage or not.
- b. To delete unnecessary pages.
- c. To determine which pages should remain in main memory and which should be swapped.
- d. To show which pages are most active/inactive

Answer: a. To determine whether it need to be rewritten to secondary storage or not.

Question 26

The 4 memory allocation schemes are being introduce to remove _____ of storing _____ contiguously.

Select one:

- a. compaction, programs
- b. restriction, programs
- c. starvation, memory
- d. deadlock, memory

Answer: b. restriction, programs

Question 27

OS can avoid deadlock if it knows in advance the sequence of process in each request.

Select one:

- True
- False

Answer: True

Question 28

What is the goal of Job Scheduler in OS?

Select one:

- a. To put jobs in a sequence.
- b. To select several jobs to run consecutively.

- c. To select several consecutive jobs with a great deal of computation.
- d. To keep most components of the computer system busy most of the time

Answer: d. To keep most components of the computer system busy most of the time

Question 29

_____ is the result of conservative allocation of resources in which a job is prevented from executing because it is kept waiting for resources that never become available.

Select one:

- a. Deadlock
- b. Starvation

Answer: b. Starvation

Question 30

Virtual memory makes it possible to swap pages between main memory and secondary storage.

Select one:

- True
- False

Answer: True

Question 31

This scheme loads only a part of the program into memory for processing and job is divided into equally sized pages that initially reside in secondary storage.

Select one:

- a. Paged memory allocation
- b. Demand paging allocation
- c. Segmented allocation
- d. Single contiguous

Answer: b. Demand paging allocation

Question 32

Process Scheduler is the low-level scheduler that assigns the CPU to execute the jobs placed in the queue.

Select one:

- True
- False

Answer: True

Question 33

FIFO swaps a page that has been in the memory shortest and LRU swaps a page that is most recently accessed.

Select one:

- True
- False

Answer: False

Question 34

In the transition from HOLD to READY mode, _____ checks availability of main memory and any requested devices.

Select one:

- a. Scheduling policy
- b. Process Scheduler
- c. Job Scheduler
- d. Scheduling algorithm

Answer: c. Job Scheduler

Question 35

LRU swaps out the pages that show the least amount of recent activity based on the _____ that these pages are the least likely to be used again in the immediate future.

Select one:

- a. duration
- b. current situation
- c. assumption
- d. number of frames

Answer: c. assumption

Question 36

What advancement has been introduced during 3rd generation computers?

Select one:

- a. Job scheduling
- b. COBOL & FORTRAN
- c. Virtual memory
- d. Time sharing and integrated circuits

Answer: d. Time sharing and integrated circuits

Question 37

One _____ holds one page of job instruction and fits into one page frame of memory.

Select one:

- a. block
- b. sector
- c. allocation
- d. frame

Answer: b. sector

Question 38

The progress of job execution is noted in _____ and is updated as the job progresses from beginning to termination.

Select one:

- a. JMT
- b. PMT
- c. PCB
- d. SMT

Answer: c. PCB

Question 39

The Process Scheduler takes over after a job has been placed in the _____ queue by the Job Scheduler.

Select one:

- a. ready
- b. resource
- c. scheduling
- d. waiting

Answer: a. ready

Question 40

What is the purpose of having Memory Map Table in Segmented/Demand Paged memory allocation scheme?

Select one:

- a. Lists details about each segment (one for each job).
- b. Monitors the allocation of page frames in main memory (one for the whole system).
- c. Lists every job in process (one for the whole system)
- d. Lists details about every page (one for each segment).

Answer: b. Monitors the allocation of page frames in main memory (one for the whole system).